

Employee Table

E-Id	E-Name	E-Add	E-Sal
1	Ansh	Bhopal	10,000
2	Shubh	Indore	20,000
3	Golu	Ujjain	16,000
4	Vicky	Ratlam	14,000
5	Ram	Jaipur	17,500
6	Prachi	Ajmer	20,500

① Find 2nd MAX SALARY ?

→ Select * from Employee where Salary =
(Select Max(E-Sal) from Employee where
E-Sal < (Select Max(E-Sal) from employee));

2) Print Id from 2 to 5 id

→ Select * from Employee where E-Id >= 2 AND
E-Id <= 5 ;

3) Print Record starts from 'a' ?

→ Select * from Employee where E-Name like 'a%' ;

4) Print record ends with a ?

→ Select * from Employee where E-Name like '%a' ;

5) Print All Record in which employee Salary is 16,000 , 20,500 , 14,000 , 10,000

→ Select * from Employee where E-Sal In (16,000 , 20,500 , 14,000 , 10,000) ;

6) Create a New table with the help of existing table and structure

→ Create Table emp-1 as Select * from Employee;

7) Find all employee Records whose Salary b/w 10,000 to 16,000 ?

Ans Select * from employee where E-Sal between 10,000 AND 16,000

or

-> Select * from Employee where E-Sal >= 10,000
AND E-Sal <= 16,000 ;

8) Write a Statement to delete an Record ?

delete from Employee where E-Id = 3 ;

9) Write a Statement to delete more than One record in Single query

-> delete from Employee where E-Id In (3, 5, 6) ;

10) Insert Record into Employee table ?

-> INSERT into Employee (E-Id, E-Name, E-Add, E-Sal)
Values (7, "Suresh", "Jamshedpur", 21,000) ;

11) Update Record name of E-Id = 1

-> Update Employee Set E-name = "Anshul" where
E-Id = 1 ;

12) Create Table Query

A-> Create Table Employee (E-Id int Primary Key,

E-name Varchar (10), E-Add Varchar (50),
E-Sal Int(30) ;

13) Foreign Key Concept

Department Table

d-Id	d-name	E-Id
1	IT	2
2	Mech	1
3	EC	3
4	Civil	5

→ Create Table Department (d-Id int Primary Key, d-name Varchar (20), E-Id int Foreign Key (E-Id) references Employee(E-Id))

Or (Cascading Concept)

→ Create Table Department (d-Id int Primary Key, d-name Varchar (20), E-Id int Foreign Key (E-Id) references Employee(E-Id) On delete Cascade On Update Cascade) ;

14) Query for Inner Join b/w employee table and Department Table

→ Select * from Employee e Join department d
On e.E-Id = d.E-Id ;

or

→ Select e.E-name, e.E-Sal, d.d-name from Employee e
Inner Join department d
On e.E-Id = d.E-Id ;

15) Left Join Query b/w employ and department table

→ Select * from Employee e
Left JOIN Department d
On e.E-Id = d.E-Id ;

or

→ Select e.E-name, e.E-Sal, d.d-name from Employee e
Left Join Department d
On e.E-Id = d.E-Id ;

17) Right Join Query b/w employee and department Table

→ Select * from Employee e

Right Join Department d
on e.E-Id = d.E-Id;

or

→ Select e.E-name, e.E-sal, d.d-name from
Employee e

Right Join Department d
on e.E-Id = d.E-Id ;

18) Unmatched data (Outer Join) b/w employee and
department Table

→ (Select * from Employee e
Left Join Department d
on e.E-Id = d.E-Id
Where e.E-Id is null) Union

(Select * from Employee e
Right Join Department d
on e.E-Id = d.E-Id
Where d.E-Id is null)

19) Full Outer JOIN b/w employ and Department Table

→ (Select * from Employee e
Left Join Department d
On e.E-Id = d.E-Id)

Union

(Select * from employee e
Right JOIN Department d
On e.E-Id = d.E-Id)

20) 5th highest Salary from Employee Table

→ Select ~~Salary~~ E-Sal from Employee Order by E-Sal desc
limit 4, 1 ;

21) 5th lowest Salary from Employee Table

Select E-Sal from Employee Order by E-Sal
limit 1 offset 4

22) Count a no. of repeated Salary of group by

→ Select ~~Salary~~, E-Sal, Count (E-Sal) from Employee
group by E-Sal ;

21 5 highest salary from Employee Table

→ Select * from Employee
Order by e.sal desc
Limit 5 ;

22> Salary between 10,000 to 20,000 Using
'Between' and Using Condition

Using Between

→ Select * from Employee where e.sal
Between 10,000 AND 20,000 ;

Using Condition

→ Select * from Employee where
e.sal $\geq$ 10,000 AND
e.sal $\leq$ 20,000 ;

23> JOIN Two Table employee and department
and Print Salary blw 10,000 to 20,000

→ Select e.e-name, e.e-sal, d.d-name from
Employee e JOIN Department d
On e.e-Id = d.e-Id
where e.e-sal Between 10,000 AND 20,000 ;

On

Select e.e-name, e.e-sal, d.d-name from Employee e
JOIN Department d
ON e.e-Id = d.e-Id
where e.e-sal $\geq$ 10,000 AND
e.e-sal $\leq$ 20,000 ;

24) Create Index On employee Table in Salary Column

→ Create Index index-salary
On Employee (e-sal) ;

25) Query for Distinct record of Salary in Employee Table

→ Select distinct e-sal from employee

26) Add Column in table

→ Alter Table Employee add (mobile int)

27 Modify Column in table

→ Alter Table ~~de~~ employee Modify Mobile
Varchar (10) ;

28) Delete Column from Table

→ Alter Table Employee drop mobile ;

29) Join Three Tables Query Join Employee Table, Department Table with Project Table.

→ Create Table Project (P-Id int Primary Key, P-name Varchar(10), E-Id Foreign Key (E-Id) references Employee (E-Id)) ;

PROJECT Table :-

P-Id	P-name	E-Id
1	BanBing	3
2	Egov	1
3	hospital-many	4
4	ORS	2

Select * from (Employee e Join Department d
on e.E-Id = d.E-Id) Join Project P
on P.E-Id = e.E-Id) ;

30) Self JOIN b/w Employee and ~~Department~~ Table

→ Select * from Employee e1, Employee e2
where e1.E-Id = e2.E-Id ;

31) Equi Join b/w Employee and Department Table

→ Select * from Employee e, Department d
where e.E-Id = d.E-Id ;

32) Create View for Employee Table

→ Create View emp-View

As

Select E-name, E-Sal

from

Employee ;

Select * from emp-View ;

33) TRIGGER Example

→ Create table Passenger (Name Varchar (20),
id int (10), address Varchar (20), charges int (10),
Primary Key (id)) ;

Insert Into Passenger Values ("Ansh", 1, "Bhopal",
10,000) ;

Insert into Passenger Values ("Shubh", 2, "Indore", 11,000);

Insert into Passenger Values ("neeraj", 3, "Ujjain", 15,000);

Select * From Passenger

Name	id	address	charges
Ansh	1	Bhopal	10,000
Shubh	2	Indore	11,000
neeraj	3	Ujjain	15,000
PASSENGER TABLE			

Create TRIGGER flight

before Insert

On Passenger

for each row

Set new.charges = new.charges - 2000 ;

insert into passenger values ("Amit", 4, "Jaipur", 10,000);

Select * from Passenger

name	id	address	charges
Amit	4	Jaipur	8000

THEORY JDBC and SQL

1) What is JDBC ?

Ans JDBC API is a Set of interfaces to access RDBMS is Called JDBC API

The JDBC API is a Set of classes and Interfaces. It is Used to access tabular data stored in a RDBMS (Relational Database Management System).

2) What is RDBMS ?

An application Can not be developed without a database. The database is Used to Keep User data into tabular format. A table Consist of Rows and Columns. Tools Man